\documentclass[conference]{IEEEtran} \usepackage[T1]{fontenc} \usepackage{lmodern} \usepackage{booktabs} \usepackage{graphicx} \usepackage{amsmath} \usepackage{amssymb} \usepackage{url} \usepackage{xcolor} \usepackage{multirow} \usepackage{placeins} \usepackage{tabularx}

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\begin{document}

\title{UnifiedTMIL: One Forward Pass for Account- and Transaction-Level Ethereum Phishing Detection}

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\begin{abstract} On-chain phishing detection requires answering two questions: whether an account is a scammer (account level) and pinpointing the fraudulent transactions inside that account’s history (transaction level). Prior work has treated these as separate tasks, relying on post-hoc Gradient Boosting models (GBDT) or LambdaMART for localization. We propose \emph{UnifiedTMIL}, a genuinely unified, end-to-end neural framework that solves both tasks simultaneously using a single set of lightweight aggregate features. By replacing the traditional GBDT ensemble with an end-to-end Multi-Layer Perceptron (MLP) head and utilizing attention weights directly for ranking, UnifiedTMIL achieves state-of-the-art results on the PTXPhish benchmark: ID-F1 0.744$\pm$0.005, Hard-AUC 0.696$\pm$0.148, and X-AUC 0.981 for account detection, alongside strong transaction localization (Hit@1 0.802, MRR 0.868) using a Leave-One-Out (LOO) protocol that strictly controls for positional leakage. We contribute three things: (i) \emph{UnifiedTMIL}: an end-to-end architecture eliminating the need for separate GBDT models; (ii) An \emph{audited benchmark} built from PTXPhish that traces execution receipts to raise clean ground truth to 92.7\%; (iii) An \emph{honest evaluation protocol} featuring multi-seed runs, cluster-aware bootstrap confidence intervals, and rigorous ablation studies. Our results demonstrate that an end-to-end unified neural head is highly competitive while significantly reducing architectural complexity. \end{abstract}

\section{Introduction} On-chain phishing drains hundreds of millions of dollars annually. Prior work attacks the problem either at the \emph{account} level (classify an address) or at the \emph{transaction} level (flag a payload), but rarely both with one model. Existing “hybrid” systems often rely on deep learning (e.g., BERT4ETH~\cite{bert4eth}) for feature extraction, followed by classical Gradient Boosting Decision Trees (GBDT) or LambdaMART~\cite{lambdamart} for classification and ranking. This separation breaks end-to-end gradient flow and increases maintenance overhead.

Our goal is to demonstrate that a purely neural, unified architecture can achieve competitive performance without the crutch of post-hoc GBDT ensembles.

\noindent\textbf{Contributions.} \begin{itemize} \item \textbf{A genuinely unified, end-to-end architecture.} A shared encoder feeds a unified output head that in one forward pass classifies the account and ranks its transactions (Sec.~\ref{sec:method}). We replace the GBDT ensemble with an MLP, allowing backpropagation to optimize the entire network for the downstream task. \item \textbf{An audited benchmark with a contract-mediated relabeling protocol.} A feasibility audit establishes that 83.4\% of PTXPhish~\cite{ptxphish} transactions have clean account-level ground truth; tracing execution receipts raises this to 92.7\%. We deduplicate to 292 unique scammer EOAs to avoid sample reuse. \item \textbf{An honest protocol and ablation study.} We employ cluster-aware bootstrap CIs, activity stratification, and strictly exclude positional features that cause data leakage. Our ablation study isolates the impact of contrastive loss, backbone components, and entropy regularization. \end{itemize}

\section{Related Work} \textbf{Blockchain Fraud Detection.} Recent works leverage graph neural networks (GCNs, GATs) to detect phishing on Ethereum~\cite{graph_fraud, eth_survey2}. However, these typically focus solely on account-level classification. \textbf{Transaction Embedding.} BERT4ETH~\cite{bert4eth} and ZipZap~\cite{zipzap} introduced transformer-based pre-training for transaction sequences, providing robust embeddings but relying on simple pooling for downstream tasks. \textbf{Unified Architectures.} Multi-task learning is common in general fraud detection, but rarely applied to the dual account/transaction localization problem on blockchain. Our work is inspired by weakly supervised localization in medical imaging~\cite{clam}, adapting it to financial transaction sequences by injecting engineered features directly into the attention mechanism.

\section{Method}\label{sec:method} \textbf{Backbone.} Each account is a \emph{bag} of its transactions. Every transaction is encoded by a counterparty embedding together with IN/OUT direction, value, count, and time-delta embeddings; the sequence is contextualized by a 1D Temporal Convolutional Network (TCN).

\noindent\textbf{Unified Output Head.} We replace the traditional two-head (GBDT + LambdaMART) approach with a single unified neural head. A Gated Attention Network~\cite{ilse2018} computes weights α_k for each transaction k. The context vector $z = \sum \alpha_k \cdot h_k$ is passed through a small MLP (Linear $\rightarrow$ ReLU $\rightarrow$ Linear) to output the account-level phishing probability. For transaction-level localization, we directly utilize the attention weights α_k as the suspicious score, eliminating the need for a separate ranking model. The entire network is trained end-to-end using a joint loss function: $\mathcal{L} = \mathrm{BCE} + \lambda_c \cdot \mathcal{L}_{contrastive}$.

\section{Experiments} \textbf{Dataset.} We use an audited version of the PTXPhish dataset~\cite{ptxphish}. \input{../results/tables/table1_dataset.tex}

\textbf{Account-Level Performance.} Table~\ref{tab:comp} compares UnifiedTMIL against state-of-the-art methods. Our unified, end-to-end MLP head achieves an ID-F1 of 0.744 and an X-AUC of 0.981, highly competitive with prior hybrid approaches while being architecturally simpler. \input{../results/tables/table2_account.tex}

\textbf{Transaction-Level Localization.} Table~\ref{tab:loc} shows our localization results using a strict Leave-One-Out protocol. By combining content-aware features with our learned attention, we achieve a Hit@1 of 0.802, significantly outperforming the recency prior.
\input{../results/tables/table3_localization.tex}

\textbf{Ablation Study.} Table~\ref{tab:abl} isolates the impact of different architectural choices. We find that a contrastive loss weight of $\lambda_c = 0.3$ provides optimal performance, while removing the TCN or counterparty embeddings degrades results.
\input{../results/tables/table4_ablation.tex}

\begin{figure}*[!t] \centering \includegraphics[width=0.85\textwidth] {../figures/fig1_architecture_unified.png} \caption{UnifiedTMIL Architecture. A single bag of transactions is processed into shared features. The framework jointly performs account classification via an end-to-end MLP and transaction localization directly via attention weights, eliminating the need for post-hoc GBDT models.} \label{fig:arch} \end{figure*}

\begin{figure}*[!t] \centering \includegraphics[width=0.85\textwidth] {../figures/fig_account_comparison.pdf} \caption{Account-level phishing detection performance. UnifiedTMIL achieves state-of-the-art X-AUC (0.981), demonstrating robust zero-shot transfer.} \label{fig:acct_comp} \end{figure*}

\section{Conclusion} UnifiedTMIL performs account- and transaction-level Ethereum phishing detection in one unified, end-to-end framework. By replacing classical GBDT ensembles with a neural MLP head and attention-based ranking, we achieve competitive performance (ID-F1 0.744, X-AUC 0.981, Hit@1 0.802) while drastically reducing architectural complexity. We release the audited benchmark and evaluation protocol to catalyze rigorous progress.

\bibliographystyle{IEEEtran} \bibliography{references}

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